

Sunset Valley: A Case Study in Computational Gossip

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Abstract—Many genres of video games value the fidelity of background non-player characters (NPCs) for they create the illusion of life necessary to support a believable world. However, in some genres of games, there is often a layer of abstraction between player actions and how NPCs process and respond to those actions. In this work, we argue that a system of computational gossip, grounded in sociological and social-psychological theory, can reduce that abstraction and improve believability of NPCs by deepening an NPCs reasoning of player action. In the form of a case study, this work presents the game *Sunset Valley*: an early step towards a system of computational gossip. Then by inspecting *Sunset Valley*'s gossip model, we analyze our game by exploring several well-established sociological and social-psychological frameworks, identify which frameworks are well-represented in *Sunset Valley*, and which frameworks are not. Finally, we conclude to what extant artificial intelligence systems can be used to address the shortcomings of our system.

Index Terms—Computational gossip, sociology, social-psychology, character believability

I. INTRODUCTION

Many genres of video games value the fidelity of background non-player characters (NPCs) because these characters, their actions, and their reactions help create the illusion of life necessary to support a believable world. Yet there is often a layer of abstraction between the player's actions and how the characters inhabiting these worlds process and respond to those actions. For example, in many roleplaying games (RPGs) the player's relationships with groups of NPCs—usually called factions—may be represented as a single numerical reputation value; such that any player action towards one faction member results in an immediate change all members within the faction. While this immediacy of change may be computationally convenient, the system fails to capture the more believable act of a player's action traveling through a social network. To better capture this information travel through social networks, we argue that a system of computational gossip is a way

to emulate believable behavior for NPCs when it comes to communicating information to other NPCs.

Though gossip is often looked upon as an unfavorable act, there exists an extensive body of sociological research that treats gossip as a tool for self-expression, a catalyst for social cohesion, and an institution that enforces societal norms [1]; thus sociology treats gossip as a tool for everyday life. Yet, despite the prevalence of gossip in our society, few computational models of gossip exist within computational media. To the best of our research, we have found only two systems that treat gossip as a first class entity: *Gossip Galore* [2] and *Grapevine* [3]. Fewer still exist that meaningfully draw upon well-established sociological and social-psychological frameworks, like Goffman's dramaturgical perspective [4]; Durkheim and Rajačić's functionalism [5] [1]; Wert and Salovy's social comparison theory [6]; and Rosnow and Fine's social exchange theory [7].

As a first step towards a model of gossip that draws upon both psychological and sociological frameworks, we designed and implemented a game centered around gossip called *Sunset Valley*. In the form of a case study, this work offers a technical description of the system of gossip used in *Sunset Valley*, followed by an internal critique of our system. In our critical analysis, we explore the aforementioned frameworks in more detail and identify which are represented well and which are underrepresented. Finally, we address the limitations of our system of gossip in *Sunset Valley* and how those limitations might be overcome by extant artificial intelligence systems.

II. CASE STUDY: *Sunset Valley*

Sunset Valley is a prototype storytelling game in which the player explores a village and interacts with seven NPCs and their stories. A core feature of the game involves gossip among these NPCs, as well as gossip between the player and NPCs. The player's ability to resolve characters' quests and pursue their story lines depends on how they have acted directly

towards these characters and if the NPCs have gossiped about the player.

A. Computational Gossip in Sunset Valley

Current computational models of gossip lean predominantly on psychological theories of social interactions based on personality traits, as shown in Khosmood and Walker’s *Grapevine* system [3]. Our model builds upon this foundation; we borrow the use of the Five-Factor Model (FFM) [8]—also known as OCEAN—to shape the personalities of our NPCs along the five dimensions of Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Whereas the intended use of *Grapevine* is to generate static text conversations among characters with those same traits, our model is meant to drive the behavior of virtual characters in a 3D world. Thus, we must take into account the spatial relationships between characters in that world. Also, our model takes into account the overall relationship, or affinity, that NPCs have with the player.

Gossip in *Sunset Valley* is motivated by three primary factors: social proxemics, or the physical distance between characters as they socialize, their individual personalities as defined by OCEAN, and their overall relationship with the player. Social proxemics serves as a prerequisite for gossip, with gossip being able to occur only when two characters are at an appropriate distance from one another. In *Sunset Valley*, this distance is calculated using the Unity engine’s physics system; we use the distance of 9 units as our starting point. The system then calculates the likelihood that two characters within the appropriate distance will gossip with each other based on their personality traits and their affinity with the player. A description of this computation is shown in Table 1. In all following tables, a plus sign next to a personality dimension (e.g., Conscientiousness+) means the value of the FFM dimension is larger than 0.75. A minus sign means the value of the FFM dimension is smaller than 0.25. An affinity denoted as “extreme” is one whose value is larger than 0.75 or lower than 0.25. Otherwise, the affinity is denoted as normal. Prior research shows that individuals who exhibit high levels of Conscientiousness, Extraversion, and Neuroticism are more inclined to spread rumors when experiencing intense emotions [9] [10]. In *Sunset Valley*, we map “intense emotions” to high or low affinity values with the player. Thus, characters that are on good or bad terms with the player are more likely to gossip about them.

TABLE I
PROBABILITY OF GOSSIP INITIATION

FFM Dimension	Affinity	Probability
Conscientiousness+	Extreme	0.85
Conscientiousness+	Normal	0.2
Conscientiousness-	All	0.7
Extraversion+	All	0.7
Extraversion-	All	0.1
Neuroticism+	All	0.8
Neuroticism-	All	0.2

Gossip content in the game is expressed through a series of affinity values, which are derived from historical affinity changes of the gossip. As the player interacts with a gossiping character, any modifications in the affinity value are recorded in the character’s social history, serving as potential material for future gossip. Interactions between the player and a character elicit changes in the character’s affinity values towards the player. These values are internal to each character, rather than being some global variable in which all characters can reference from. The affinities will then be the primary motivator if a character should gossip when they are close to another character. If a character chooses to gossip, then these conversations will be regarded as the gossip content and result in the appropriate affinity change. That content will then be transmitted to all characters that are nearby the gossip.

A character’s personality traits also impact their susceptibility to gossip, with those scoring high in Extraversion and Neuroticism being more vulnerable to rumors [11]. Studies also show that individuals with high extraversion value react positively to positive content and negatively to negative content [12]. There were no significant differences between the positive or negative emotions in extraverted individuals. People of high neuroticism react favorably toward negative content. They also react more positive to everything in the supportive emotion as compared to the antagonistic ones. In *Sunset Valley*, we posit that relationships between the player and characters are positively correlated with emotions mentioned above. Specifically, positive affiliations have a propensity to elicit fervent responses, whereas negative associations tend to elicit sentiments of disgust. Thus, we shift the focus from emotions to relationships in our gossip formula. We represent these findings through higher or lower impact affinity changes in response to a character’s personality, current affinity with the player, and whether or not the gossip content elicits positive or negative affinity changes, as shown in Table 2. In this table, the value “afval” serves as a representation of the gossip content transmitted to the listener. This value is derived from the gossip’s historical affinity changes, which are accumulated through interactions with the player.

TABLE II
EFFECTS OF GOSSIP

FFM Dimension	Curr. Affinity	Pos. Change	Neg. Change
Extraversion+	Positive	afval *= 1.3	afval *= 0.7
Extraversion+	Negative	afval *= 1.3	afval *= 0.7
Extraversion-	Positive	afval *= 1	afval *= 1
Extraversion-	Negative	afval *= 1	afval *= 1
Neuroticism+	Positive	afval *= 0.8	afval *= 1.2
Neuroticism+	Negative	afval *= 0.6	afval *= 1.4
Neuroticism-	Positive	afval *= 1	afval *= 1
Neuroticism-	Negative	afval *= 1	afval *= 1

Studies show that individuals with high extraversion value in the positive range, react positively to positive content and negatively to negative content. There were no significant differences between the positive or negative emotions in extraverted individuals. People of high neuroticism value in the negative

emotion as compared with the positive emotion, react favorably toward negative content. They also reacted more positive to everything in the supportive emotion as compared to the antagonistic ones. In *Sunset Valley*, we posit that relationships between the player and characters are positively correlated with emotions mentioned above. Specifically, positive affiliations have a propensity to elicit fervent responses, whereas negative associations tend to elicit sentiments of disgust. Thus, we shift the focus from emotions to relationships in our gossip formula.

B. Analyzing Gossip in *Sunset Valley*

Given that our ultimate goal is to produce a model of gossip that is representative of modern theory, we offer a critical analysis of our system through multiple established frameworks of gossip. First, we analyze the system through the lens of strongly rooted sociological theories of gossip, including the dramaturgical and functionalist perspective. Second, we analyze the system through the lens of social-psychological theories of gossip, including social exchange theory and social comparison theory. Finally, after identifying the gaps in our model, we describe how we might improve its fidelity by incorporating suitable artificial intelligence systems and languages, like *Comme il Faut* (CiF) and A Behavior Language (ABL).

1) *Dramaturgy*: Dramaturgy in the presentation of self, as proposed by Goffman, is a prominent sociological lens for viewing gossip [4]. Goffman contends that individuals perform actions in everyday life as if they were performers on a stage, and each performance is a presentation of self such that each individual seeks to create specific impressions in the minds of others. Other researchers in the field of social psychology apply this perspective directly to the role of gossip, characterizing it as, "a strategy individuals use to manage impressions, to improve their reputation, and further their interests at the expense of others" [13].

Our system of gossip, and the gameplay surrounding it, effectively display these aspects of dramaturgy. *Sunset Valley* presents the player with dialogue options that lead to varying storylines. These dialogue options have the potential to affect the player's affinity level with surrounding NPC's. As such, the player's decision whether or not to spread gossip is largely dependent on the perceived positive or negative consequences of engagement, and the NPCs possess unique personalities that inform their propensity for gossip. In this way the choices presented in *Sunset Valley* can be regarded as a representation of dramaturgical self presentation.

2) *Functionalism*: The functionalist theory of sociology argues that there are social rules which exist beyond that which individuals control and that ultimately govern their everyday behaviors [5]. These social rules are developed and upheld through institutions—gossip included—all of which have a function in maintaining social order. Rajačić's metatheory of gossip adds that gossip is an institution that maintains social cohesion through punishment and reward of those who reject or accept social norms [1].

While characters in *Sunset Valley* possess individual traits, their actions are not governed by a larger body of knowledge and principles as Durkheim describes. A future model of gossip should work to include a universal database of rules or social facts which inform all actors within the simulation beyond the individual traits they possess. Because our system will need this database of rules and social facts, we believe that *Comme il Faut* (CiF) [14], a social AI system, fills a foundational part of our future model. Later in this section, we will explain more about CiF and its utility for a future gossip system built on top of *Sunset Valley*'s gossip model.

3) *Social Comparison and Social Exchange*: Wert and Salovy's theory of gossip as a tool for social comparison articulates multiple ways in which actors will use gossip to place themselves relative to others in society [6]. The most relevant comparisons for our purposes are those made to actors who are more or less "able, powerful or fortunate." Upward comparison or gossip about those more fortunate than oneself allows the gossipier access to information that would otherwise be inaccessible to members of their social class. Downward comparisons or gossip concerning those less fortunate than oneself allows the gossipier to elevate their social status by placing themselves above the subject of the gossip.

Social exchange theory, on the other hand, asserts that gossip is "an instrumental transaction in which A and B trade small talk about C for something in return – more gossip, status, fun, money, social control, or any material or psychological stimulus" [7]. Similarly, Rajačić's metatheory argues that the value of gossip is tied to its usefulness and rarity [1]. These ideas of status and rarity are therefore linked to the larger social hierarchy mentioned within the social comparison theory. Resulting power dynamics largely inform the response an actor has to the gossip they receive.

Both of these ideas are highly relevant in social theory surrounding real gossip, but are impossible in our current simulation due to the lack of social hierarchy. Our future model will make an effort to simulate a social hierarchy to more authentically represent the potential of gossip as a tool for social mobility.

4) *Using Comme il Faut*: If our goal is to create a believable model of gossip, we must base all representations in a larger social context that a database of social facts would provide. *Sunset Valley* currently represents gossip as it relates to the self; building a more rooted model of gossip requires a great deal more regarding social context. Additionally, this requirement of social context through a database makes CiF once again a promising part of the system as CiF was the driving AI behind *Prom Week*, a game focused around the social interactions between characters during a week leading up to their school prom. While gossip was not a first class entity within *Prom Week*, it did appear in some sense via the twitter-like system in place. CiF achieved this through having a production database of social rules and facts that were queried during each character interaction, allowing characters a deeper way of expressing themselves. We hope to use CiF to better build around gossip, implementing the already

mentioned theories within this section.

5) *Using A Behavior Language*: We mentioned earlier in this paper that *Sunset Valley* uses distance between characters to determine whether gossip should occur. While this light representation of social proxemics is suitable for the purposes of *Sunset Valley*, we believe there are more proxemics-related considerations to make, such as: the capacity for more than two characters who are near to each other to gossip; simultaneous gossip amongst all groups of characters who are near to each other; and appropriate settings in which characters might gossip. Thus, A Behavior Language (ABL), a programming language built for real-time character behaviors [15], could be leveraged to coordinate such context-dependent, multi-character actions. For instance, using a combination of ABL's joint behaviors and daemon patterns [16], our system could more efficiently determine when characters are grouped in clusters of two or more at an appropriate distance and orchestrate a unified behavior amongst all of those same characters.

III. DISCUSSION AND FUTURE WORK

In the prior sections, we identified the following needs for our model: a database of social rules and facts for character motivation; an expansion of the physical mechanics of gossip as it relates to social proxemics; and a rich representation of dynamic social hierarchy that provides the basis for both social exchange and social comparison. By using CiF and ABL, we believe the absence of these concepts in our model can be addressed. As for this section, we discuss our work thus far and our plans for future work on an improved model of computational gossip.

While not quite like a research game postmortem, as described by Horswill within his MKULTRA postmortem [17], we believe that, like Horswill, we have learned many lessons from developing *Sunset Valley*. While *Sunset Valley* could be further developed alongside our new model of gossip, we have decided to rehouse our model in an existing game upon which we can build: *Oops! All Bards*, a fantasy RPG game inspired by the medieval-fantasy genre. We base this decision on the following: CiF and ABL have already been successfully connected within *Oops! All Bards* [18]; the game's thematic trappings will help us define a social hierarchy that can provide unique opportunities for social exchange and comparison; and its implementation in 3D will be a suitable testing ground for the social proxemics part of our gossip model.

IV. CONCLUSION

In this paper, we describe a prototype of computational gossip, which we use within *Sunset Valley* as a first cut of a more comprehensive model of gossip. We did this by first describing *Sunset Valley*'s model of gossip, which uses OCEAN as its underlying gossip model. A brief explanation of how we came up with our model of gossip is given. Then, we inspected our model of gossip and discovered that it was lacking in the sense that it does not fully capture what is happening when people gossip. To try and improve the system, we looked towards sociology and social-psychology

to increase the scope of our gossip model. By looking at what sociological models *Sunset Valley* does well, like dramaturgy, and where *Sunset Valley* doesn't capture the essence of other sociological/social-psychological models, like Functionalism, Social Exchange, and Social Comparison, we hypothesize a solution through combining two AI systems, CiF and ABL. Finally, we lay out an explanation of the future system within the game *Oops! All Bards*.

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